

CANTIDAD DE MOVIMIENTO Y CHOQUES

$$\mathbf{p} = m\mathbf{v}, \quad \sum \mathbf{F} = \frac{d\mathbf{p}}{dt}$$
$$\mathbf{J} = \int \mathbf{F}dt = \Delta\mathbf{p} = \mathbf{F}_{\text{med}}\Delta t$$
$$\mathbf{r}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{M}, \quad \mathbf{P} = M\mathbf{v}_{\text{cm}}$$
$$\sum \mathbf{F}_{\text{ext}} = M\mathbf{a}_{\text{cm}}, \quad \sum \mathbf{F}_{\text{ext}} = 0 \Rightarrow \mathbf{P} = \text{cte}$$
$$m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 = m_1 \mathbf{v}'_1 + m_2 \mathbf{v}'_2$$
$$\mathbf{v}'_1 = \mathbf{v}'_2 = \frac{\mathbf{P}}{M}, \quad K_{\text{perd}} = \frac{1}{2}m_r v_{\text{rel}}^2$$
$$m_r = \frac{m_1 m_2}{m_1 + m_2}, \quad \mathbf{v}_{\text{rel}} = \mathbf{v}_A - \mathbf{v}_B$$
$$K = \frac{1}{2}Mv_{\text{cm}}^2 + K_{\text{rel}}, \quad K = \frac{p^2}{2m}$$

COHETE

$$\mu = -\frac{dM}{dt}, \quad f = \mu|v_r| = I_{\text{sp}}g_0\mu, \quad I_{\text{sp}} = \frac{|v_r|}{g_0}$$
$$M\frac{dV}{dt} = \mu|v_r| - Mg, \quad M(t^*) = \frac{\mu|v_r|}{g}$$
$$M(t) = M_0 - \mu t, \quad a(t) = \frac{\mu|v_r|}{M_0 - \mu t} - g$$
$$V(t) = V_0 + |v_r| \ln \frac{M_0}{M(t)} - gt$$
$$y(t) = y_0 + V_0 t - \frac{1}{2}gt^2 + |v_r| \left[t - \frac{M(t)}{\mu} \ln \frac{M_0}{M(t)} \right]$$
$$\Delta V = |v_r| \ln \frac{M_0}{M_f}, \quad \frac{M_0}{M_f} = \exp \left(\frac{\Delta V}{|v_r|} \right)$$
$$\Delta V_{\text{tot}} = \sum \Delta V_i, \quad \mathbf{F}_{\text{ext}} = \frac{d(M\mathbf{v})}{dt}$$

TRABAJO Y ENERGÍA

$$W = \int \mathbf{F} \cdot d\mathbf{l}, \quad W_{\text{tot}} = \Delta K, \quad \mathbf{P} = \mathbf{F} \cdot \mathbf{v}$$
$$\mathbf{F} = -\nabla U, \quad E = K + U, \quad W_{\text{no cons}} = \Delta E$$

MOMENTO ANGULAR

$$\mathbf{L}_O = \mathbf{r} \times m\mathbf{v}, \quad L = mvr \sin \varphi = mvd$$
$$\sum \tau_O = \frac{dL_O}{dt}, \quad \int \tau dt = \Delta L$$
$$h = rv_{\perp} = rv \cos \gamma = r^2 \dot{\theta} = \text{cte}, \quad \frac{dA}{dt} = \frac{h}{2}$$
$$r_P v_P = r_A v_A$$

ROTACIÓN Y GIRÓSCOPO

$$v = \omega r, \quad a_t = \alpha r, \quad a_{\text{rad}} = \omega^2 r, \quad K = \frac{1}{2}I\omega^2$$
$$I = \sum m_i r_i^2, \quad I_P = I_{\text{cm}} + Md^2$$

varilla (cm)	$ML^2/12$	aro	MR^2
varilla (ext)	$ML^2/3$	disco	$MR^2/2$
esfera	$2MR^2/5$	esf. hueca	$2MR^2/3$

$$\tau = rF \sin \varphi, \quad \sum \tau_z = I\alpha_z, \quad L = I\omega$$
$$\sum \tau_{\text{ext}} = \frac{dL}{dt}$$
$$\sum \tau_{\text{ext}} = 0 \Rightarrow \sum I_i \omega_i = \text{cte}$$
$$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2, \quad v_{\text{cm}} = R\omega$$
$$\Delta L = \tau \Delta t, \quad \Omega = \frac{\tau}{L} = \frac{Mgr}{I\omega}$$

GRAVITACIÓN

$$\mathbf{F}_g = -\frac{GMm}{r^2}\hat{\mathbf{r}}, \quad U = -\frac{GMm}{r}$$
$$g_0 = \frac{GM}{R^2} = \frac{\mu}{R^2}, \quad g(h) = g_0 \left(\frac{R}{R+h} \right)^2$$
$$v_c = \sqrt{\frac{\mu}{r}}, \quad v_{\text{esc}} = \sqrt{\frac{2\mu}{r}} = \sqrt{2}v_c, \quad T = \frac{2\pi r}{v_c}$$
$$\Delta E_{R \rightarrow r} = \mu m \left(\frac{1}{R} - \frac{1}{r} \right)$$
$$\Delta E_{r_1 \rightarrow r_2} = \frac{\mu m}{2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

DOS CUERPOS

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r}, \quad \mu = G(m_1 + m_2) \approx Gm_1$$
$$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2, \quad \mathbf{r}_1 = \frac{m_2}{M}\mathbf{r}, \quad \mathbf{r}_2 = -\frac{m_1}{M}\mathbf{r}$$
$$E = \frac{1}{2}m\dot{\mathbf{r}}^2 + \frac{L^2}{2mr^2} + U(r), \quad U_{\text{ef}} = U + \frac{L^2}{2mr^2}$$

ÓRBITAS

$$r = \frac{p}{1 + e \cos \nu}, \quad p = \frac{h^2}{\mu} = a(1 - e^2)$$
$$\cos \nu = \frac{1}{e} \left(\frac{p}{r} - 1 \right), \quad e \sin \nu = \frac{h v_r}{\mu}$$
$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}, \quad v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$$
$$e = \sqrt{1 + \frac{2\varepsilon h^2}{\mu^2}} = \frac{r_A - r_P}{r_A + r_P} = \frac{r_P v_P^2}{\mu} - 1$$
$$e^2 = \left(\frac{p}{r} - 1 \right)^2 + \left(\frac{h v_r}{\mu} \right)^2$$
$$e = \frac{r_2 - r_1}{r_1 \cos \nu_1 - r_2 \cos \nu_2}$$
$$h = rv \cos \gamma = r_P v_P = \sqrt{\mu p} = \sqrt{\mu a(1 - e^2)}$$
$$r_P = a(1 - e) = \frac{p}{1 + e}, \quad r_A = a(1 + e) = \frac{p}{1 - e}$$
$$a = \frac{r_P + r_A}{2}, \quad b = a\sqrt{1 - e^2} = \sqrt{r_P r_A}$$
$$\frac{1}{2}v_P^2 - \frac{\mu}{r_P} = \frac{1}{2}v_A^2 - \frac{\mu}{r_A}$$
$$v_P = \sqrt{\frac{2\mu r_A}{r_P(r_P + r_A)}}, \quad v_A = \frac{r_P}{r_A}v_P$$
$$v_P = \sqrt{\frac{\mu}{a} \frac{1 + e}{1 - e}}, \quad v_A = \sqrt{\frac{\mu}{a} \frac{1 - e}{1 + e}}$$
$$v_{\perp} = \frac{h}{r} = \frac{\mu}{h}(1 + e \cos \nu), \quad v_r = \frac{\mu}{h}e \sin \nu$$
$$\tan \gamma = \frac{v_r}{v_{\perp}} = \frac{e \sin \nu}{1 + e \cos \nu}$$

HIPÉRBOLA Y ESCAPE

$$e = 1: \quad \varepsilon = 0, \quad v = v_{\text{esc}} = \sqrt{\frac{2\mu}{r}}$$
$$\varepsilon = \frac{v_{\infty}^2}{2} = \frac{\mu}{2a}, \quad v^2 = v_{\text{esc}}^2 + v_{\infty}^2, \quad C_3 = v_{\infty}^2$$
$$a = \frac{p}{e^2 - 1} = \frac{\mu}{v_{\infty}^2}, \quad r_P = a(e - 1)$$
$$e = 1 + \frac{r_P v_{\infty}^2}{\mu}$$
$$\nu_{\infty} = \arccos \left(-\frac{1}{e} \right), \quad \delta = 2 \arcsin \left(\frac{1}{e} \right)$$
$$\Delta v_{\text{esc}} = \sqrt{\frac{2\mu}{r}} - v$$

KEPLER

$$T = 2\pi \sqrt{\frac{a^3}{\mu}} = \frac{2\pi ab}{h}, \quad n = \frac{2\pi}{T} = \sqrt{\frac{\mu}{a^3}}$$
$$\tan \frac{E}{2} = \sqrt{\frac{1 - e}{1 + e}} \tan \frac{\nu}{2}, \quad \cos E = \frac{e + \cos \nu}{1 + e \cos \nu}$$
$$M_e = E - e \sin E = n(t - t_P)$$
$$r = a(1 - e \cos E)$$
$$T_{\text{sid}} = 24 \text{ h} \cdot \frac{360^\circ}{360,986^\circ} = 23,934 \text{ h}$$

MANIOBRAS

$$v_P = \sqrt{\frac{2\mu r_2}{r_1(r_1 + r_2)}}, \quad v_A = \frac{r_1}{r_2}v_P$$
$$\Delta v_1 = v_P - v_{c1}, \quad \Delta v_2 = v_{c2} - v_A, \quad v_{ci} = \sqrt{\frac{\mu}{r_i}}$$
$$a_t = \frac{r_1 + r_2}{2}, \quad t_v = \frac{T_t}{2} = \pi \sqrt{\frac{a_t^3}{\mu}}$$
$$\varphi = 180^\circ - n_2 t_v, \quad n_2 = 360^\circ/T_2$$
$$\frac{T'}{T} = 1 - \frac{\Delta \varphi}{360^\circ}, \quad a' = r \left(\frac{T'}{T} \right)^{2/3}$$
$$r_{P'} = 2a' - r$$

ELEMENTOS ORBITALES

$$(\mathbf{r}, \mathbf{v}) \leftrightarrow (a, e, i, \Omega, \omega, \nu_0)$$
$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{I} & \hat{J} & \hat{K} \\ A_I & A_J & A_K \\ B_I & B_J & B_K \end{vmatrix}$$
$$= (A_J B_K - A_K B_J)\hat{I} - (A_I B_K - A_K B_I)\hat{J} + (A_I B_J - A_J B_I)\hat{K}$$
$$|\mathbf{A} \times \mathbf{B}| = AB \sin \alpha, \quad \mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$$
$$\hat{I} \times \hat{J} = \hat{K}, \quad \hat{J} \times \hat{K} = \hat{I}, \quad \hat{K} \times \hat{I} = \hat{J}$$
$$\mathbf{A} \cdot \mathbf{B} = A_I B_I + A_J B_J + A_K B_K = AB \cos \alpha$$
$$\mathbf{r} \cdot \mathbf{v} = rv_r = rv \sin \gamma$$
$$\mathbf{h} = \mathbf{r} \times \mathbf{v}, \quad \mathbf{n} = \hat{\mathbf{K}} \times \mathbf{h} = (-h_J, h_I, 0)$$
$$e = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v} \right]$$
$$p = \frac{h^2}{\mu}, \quad e = |e|, \quad a = \frac{p}{1 - e^2} = -\frac{\mu}{2\varepsilon}$$
$$\cos i = \frac{h_K}{h}$$
$$\cos \Omega = \frac{n_I}{n} \quad (n_J < 0 \Rightarrow \Omega > 180^\circ)$$
$$\cos \omega = \frac{\mathbf{n} \cdot \mathbf{e}}{ne} \quad (e_K < 0 \Rightarrow \omega > 180^\circ)$$
$$\cos \nu_0 = \frac{\mathbf{e} \cdot \mathbf{r}}{er} \quad (\mathbf{r} \cdot \mathbf{v} < 0 \Rightarrow \nu_0 > 180^\circ)$$
$$\Pi = \Omega + \omega \quad (i = 0)$$
$$u_0 = \omega + \nu_0 \quad (e = 0)$$
$$l_0 = \Omega + \omega + \nu_0 \quad (i = e = 0)$$
$$\cos \Pi = \frac{e_I}{e} \quad (e_J < 0 \Rightarrow \Pi > 180^\circ)$$
$$\cos u_0 = \frac{\mathbf{n} \cdot \mathbf{r}}{nr} \quad (r_K < 0 \Rightarrow u_0 > 180^\circ)$$
$$\cos l_0 = \frac{r_I}{r} \quad (r_J < 0 \Rightarrow l_0 > 180^\circ)$$
$$\mathbf{r} = r \cos \nu \hat{P} + r \sin \nu \hat{Q}$$
$$\mathbf{v} = \frac{\mu}{h} \left[-\sin \nu \hat{P} + (e + \cos \nu) \hat{Q} \right]$$

CONSTANTES

G	$6,674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$
μ_T, R_T	$398\,600 \text{ km}^3/\text{s}^2, 6378 \text{ km}$
g_0	$9,80665 \text{ m/s}^2$
μ_{Luna}	$4903 \text{ km}^3/\text{s}^2, d_{\text{T-L}} \approx 384\,400 \text{ km}$
μ_{Sol}	$1,327 \times 10^{11} \text{ km}^3/\text{s}^2$
1 UA	$149,6 \times 10^6 \text{ km}$

MARCOS NO INERCIALES

$$\boldsymbol{v} = \dot{r}\hat{\boldsymbol{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}}$$
$$\boldsymbol{a} = \left(\ddot{r} - r\dot{\theta}^2\right)\hat{\boldsymbol{r}} + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\hat{\boldsymbol{\theta}}$$
$$m\boldsymbol{a}' = \boldsymbol{F} - m\boldsymbol{A}$$
$$m\boldsymbol{a}_{\text{rel}} = \boldsymbol{F} - 2m\boldsymbol{\Omega} \times \boldsymbol{v}_{\text{rel}} - m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \boldsymbol{r}) - m\dot{\boldsymbol{\Omega}} \times \boldsymbol{r}$$

ESFERA DE INFLUENCIA Y CÓNICAS PARCHEADAS

$$r_{\text{SOI}} = R\left(\frac{m_p}{m_s}\right)^{2/5}, \quad r_{\text{SOI,T}} \approx 925\,000 \text{ km}$$
$$\boldsymbol{v}_{\infty} = \boldsymbol{V}_{\text{nave}} - \boldsymbol{V}_{\text{planeta}}$$
$$v_P = \sqrt{v_{\infty}^2 + \frac{2\mu}{r_P}}, \quad v_c = \sqrt{\mu/r_P}$$
$$e = 1 + \frac{r_P v_{\infty}^2}{\mu}$$
$$\Delta v = v_P - v_c = v_c \left(\sqrt{2 + (v_{\infty}/v_c)^2} - 1\right)$$

COEFICIENTES DE LAGRANGE

$$\boldsymbol{r} = f\boldsymbol{r}_0 + g\boldsymbol{v}_0, \quad \boldsymbol{v} = \dot{f}\boldsymbol{r}_0 + \dot{g}\boldsymbol{v}_0$$
$$f\dot{g} - \dot{f}g = 1$$
$$f = 1 - \frac{\mu r}{h^2}(1 - \cos \Delta\nu), \quad g = \frac{rr_0}{h} \sin \Delta\nu$$
$$\dot{g} = 1 - \frac{\mu r_0}{h^2}(1 - \cos \Delta\nu)$$

TRES CUERPOS RESTRINGIDO

$$\Omega = \sqrt{\mu/r_{12}^3}, \quad \pi_i = \frac{m_i}{m_1 + m_2}$$
$$x_1 = -\pi_2 r_{12}, \quad x_2 = \pi_1 r_{12}$$
$$\ddot{x} - 2\Omega\dot{y} - \Omega^2 x = -\frac{\mu_1}{r_1^3}(x + \pi_2 r_{12}) - \frac{\mu_2}{r_2^3}(x - \pi_1 r_{12})$$
$$\ddot{y} + 2\Omega\dot{x} - \Omega^2 y = -\left(\frac{\mu_1}{r_1^3} + \frac{\mu_2}{r_2^3}\right)y$$
$$L_{4,5} : \left(\frac{r_{12}}{2} - \pi_2 r_{12}, \pm \frac{\sqrt{3}}{2}r_{12}\right)$$
$$r_{\text{Hill}} = r_{12}\left(\frac{m_2}{3m_1}\right)^{1/3}, \quad \frac{m_1}{m_2} + \frac{m_2}{m_1} \geq 25$$
$$C = \frac{v^2}{2} - \frac{\Omega^2(x^2 + y^2)}{2} - \frac{\mu_1}{r_1} - \frac{\mu_2}{r_2}$$

CUERPO RÍGIDO EN 3D

$$\boldsymbol{v}_B = \boldsymbol{v}_A + \boldsymbol{\omega} \times \boldsymbol{r}_{B/A}$$
$$\boldsymbol{a}_B = \boldsymbol{a}_A + \boldsymbol{\alpha} \times \boldsymbol{r}_{B/A} + \boldsymbol{\omega} \times \left(\boldsymbol{\omega} \times \boldsymbol{r}_{B/A}\right)$$
$$\left(\dot{\boldsymbol{Q}}\right)_{\text{fijo}} = \left(\dot{\boldsymbol{Q}}\right)_{\text{rot}} + \boldsymbol{\Omega} \times \boldsymbol{Q}$$
$$\begin{pmatrix} H_x \\ H_y \\ H_z \end{pmatrix} = \begin{pmatrix} I_x & -I_{xy} & -I_{xz} \\ -I_{xy} & I_y & -I_{yz} \\ -I_{xz} & -I_{yz} & I_z \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$
$$\boldsymbol{H}_O = \bar{\boldsymbol{r}} \times m\bar{\boldsymbol{v}} + \boldsymbol{H}_G, \quad T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\boldsymbol{\omega} \cdot \boldsymbol{H}_G$$
$$\sum M_x = I_x \dot{\omega}_x - (I_y - I_z)\omega_y \omega_z$$
$$\sum M_y = I_y \dot{\omega}_y - (I_z - I_x)\omega_z \omega_x$$
$$\sum M_z = I_z \dot{\omega}_z - (I_x - I_y)\omega_x \omega_y$$
$$\sum M_O = \dot{\varphi} \sin \theta \left[I'\dot{\psi} + (I' - I)\dot{\varphi} \cos \theta\right] \hat{f}$$
$$\theta = 90^\circ : \quad \sum M_O = I'\dot{\varphi}\dot{\psi}\hat{f}$$
$$\dot{\varphi} = H/I, \quad \tan \gamma = \frac{I}{I'} \tan \theta$$